

Lecture 1

September 24, 2025

1 The Two Faces of Quantum Computation

There are two complementary aspects of quantum computation:

1. The theoretical model of quantum computation.

- An idealized, consistent, plausible description of reality.
- “Implementation agnostic.”
- A testbed for algorithm design and proving impossibility results.

2. Implementations of quantum computation.

- Superconducting qubits, photonic systems, neutral atoms, etc.
- Models in which we hope to simulate the idealized model.
- Not the main focus of this class, but pertinent to the implementation and manipulation of quantum information.

An example of quantum-computing advantage.

2 Notation

We express vectors over $\mathbb{C}^d$ as

$$|v\rangle = \begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{d-1} \end{pmatrix}.$$

Because of its prevalence, we write

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{d-1}|d-1\rangle.$$

The complex conjugate is denoted by $(\cdot)^*$.

For vectors,

$$\langle v| = \begin{pmatrix} v_0^* & v_1^* & \cdots & v_{d-1}^* \end{pmatrix}, \quad \langle v|v\rangle = \sum_{j=0}^{d-1} |v_j|^2.$$

Questions for class.

1. What is $\langle 1|1\rangle$?
2. Let $M = |1\rangle\langle 1|$ (short form $|1\rangle\langle 1|$). For unit vectors $|v\rangle$, what is $M|v\rangle$?

3 Axiom 1: State Description

The state of exactly one qubit can be expressed as a vector in $\mathbb{C}^2$ of unit norm:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Define

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Note that

$$\langle 0 | 1 \rangle = 0.$$

These form an orthonormal basis for $\mathbb{C}^2$ and are possible states for a qubit.

If the state of a qubit is either $|0\rangle$ or $|1\rangle$, we call it *classical*. For example,

$$|0\rangle + |1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

(up to normalization) is a non-classical, i.e. “quantum,” possible state.

4 Axiom 2: Transformation of a Qubit

A quantum transformation is represented by a unitary matrix U :

$$|\psi\rangle \mapsto U|\psi\rangle.$$

For a unitary matrix,

$$U^\dagger U = I,$$

and consequently a unit vector remains a unit vector:

$$\|U|\psi\rangle\| = \||\psi\rangle\|.$$

5 Single-Qubit Gates

Important examples include:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{“bit flip,”}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{“phase flip,”}$$

and the Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{“Hadamard.”}$$

6 Axiom 3: Measurement / Born's Rule

Given a quantum state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

we can measure the quantum state.

The probability of observing 0 is

$$\Pr(0) = |\alpha|^2,$$

in which case the state collapses to $|0\rangle$.

The probability of observing 1 is

$$\Pr(1) = |\beta|^2,$$

in which case the state collapses to $|1\rangle$.

The classical value “0” or “1” is output.

6.1 Repeated Measurement

If the state is $|0\rangle$ or $|1\rangle$, measurement does not change the state. Hence $|0\rangle$ and $|1\rangle$ behave as classical values: measurement / observation does not change the state.

7 Global Phase

Consider the states

$$|\psi\rangle \quad \text{and} \quad e^{i\theta}|\psi\rangle.$$

These states cannot be distinguished by measurement. For example,

$$\Pr[e^{i\theta}|\psi\rangle \text{ collapsing to } |j\rangle] = |\langle j|e^{i\theta}|\psi\rangle|^2 = |e^{i\theta}\langle j||\psi\rangle|^2 = |\langle j||\psi\rangle|^2.$$

Thus $e^{i\theta}$ is called the *global phase*, and quantum states form an equivalence class

$$|\psi\rangle \sim e^{i\theta}|\psi\rangle.$$

The set of physical pure states is therefore the corresponding quotient space (projective Hilbert space).

8 Axiom 4: Initialization

We can initialize a qubit as

$$|\psi_0\rangle = |0\rangle.$$

These axioms already imply a perfect random-number generator.

1. Initialize the qubit as $|\psi_0\rangle = |0\rangle$.

2. Apply the Hadamard transform:

$$|\psi_1\rangle = H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

3. Measure the qubit:

$$\Pr(0) = \frac{1}{2}, \quad \Pr(1) = \frac{1}{2}.$$

9 Multiple Qubits

Recall the notion of tensor products.

For matrices A and B ,

$$A \otimes B$$

is the Kronecker (tensor) product. If $A = (a_{ij})$ and $B = (b_{kl})$, then

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

9.1 Properties of the Tensor Product

For compatible matrices,

1. Associativity:

$$(A \otimes B) \otimes C = A \otimes (B \otimes C).$$

2. Linearity:

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

3. Transpose:

$$(A \otimes B)^T = A^T \otimes B^T.$$

4. Conjugate transpose:

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger.$$

5. When invertible,

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

We can also take tensor products of vectors, since vectors are 1-D matrices.

9.2 Exercises

1. What is $|0\rangle \otimes |1\rangle$?
2. What is $|0\rangle \otimes |0\rangle \otimes |1\rangle$?

10 Computational-Basis Notation

We use the shorthand

$$|0\rangle|1\rangle|0\rangle = |010\rangle.$$

More generally,

$$|x_1x_2\cdots x_n\rangle, \quad x_i \in \{0, 1\},$$

denotes the computational-basis state whose bit string is $x_1x_2\cdots x_n$.

For $j \in \{0, \dots, 2^n - 1\}$, we write

$$|j\rangle$$

for the basis vector corresponding to the binary representation of j .

Equivalently,

$$|j\rangle = |j_1j_2\cdots j_n\rangle,$$

where $j_1j_2\cdots j_n$ is the n -bit binary representation of j . These basis states are orthogonal:

$$\langle j | k \rangle = \delta_{jk}.$$

For $|j\rangle \in \mathbb{C}^d$,

$$|j\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix},$$

with the 1 in the j th position (up to the indexing convention).

11 Bringing Multiple Qubits Together

Suppose qubit A is in state $|\psi_A\rangle$ and qubit B is in state $|\psi_B\rangle$. We describe the state of the two qubits by

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle.$$

We need axioms to describe the composite system.

1. If qubits do not interact, the description should reproduce the statistics of the individual qubits.
2. There must be a method for entangling the qubits; i.e. for any desired state there should exist a method of generating that state.

12 Axioms for n Qubits

1. The state of an n -qubit system is a unit vector in

$$(\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n},$$

the n -fold tensor-product Hilbert space.

Every quantum state is therefore of the form

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \sum_x |\alpha_x|^2 = 1.$$

2. For any unitary $U \in \mathbb{C}^{2^n \times 2^n}$, we can transform

$$|\psi\rangle \mapsto U|\psi\rangle.$$

In particular, products of local and two-qubit gates can generate general quantum transformations. The lecture notes emphasize that allowing gates acting on adjacent qubits is sufficient for the intended construction.

12.1 Classical States and Superposition

A state

$$|\psi\rangle = \sum_x \alpha_x |x\rangle$$

is *classical* if all amplitudes are zero except one (whose magnitude is 1).

A state is in *superposition* if there are multiple non-zero amplitudes.

13 Local Gates and Two-Qubit Gates

A unitary in $\mathbb{C}^{2 \times 2}$ is a single-qubit unitary. In particular, $H \in \mathbb{C}^{2 \times 2}$ is a single-qubit unitary.

A unitary in $\mathbb{C}^{4 \times 4}$ is a two-qubit unitary.

A circuit of the form

$$I \otimes \cdots \otimes I \otimes H \otimes I \otimes \cdots \otimes I$$

acts on one qubit, while a two-qubit gate acts on a pair of qubits.

More generally, a product of such local gates and two-qubit gates generates a unitary transformation on the full n -qubit space.

13.1 Linearity

Suppose

$$U = I \otimes \cdots \otimes I \otimes H \otimes I \otimes \cdots \otimes I.$$

For a computational-basis state

$$|x_1 x_2 \cdots x_n\rangle,$$

the action factors according to the qubit on which H acts. By linearity, for

$$|\psi\rangle = \sum_{x_1, \dots, x_n} \alpha_{x_1 \dots x_n} |x_1 \cdots x_n\rangle,$$

we obtain

$$U|\psi\rangle = \sum_{x_1, \dots, x_n} \alpha_{x_1 \dots x_n} U|x_1 \cdots x_n\rangle.$$

14 Measurement of All Qubits

The n -qubit version of Born's rule is:

$$|\psi\rangle = \sum_{j=0}^{2^n-1} \alpha_j |j\rangle \implies \Pr(j) = |\alpha_j|^2.$$

Upon observing outcome j , the state collapses to

$$|j\rangle.$$

The problem set will introduce how to measure arbitrary single qubits.

14.1 Measurement of the First Qubit

For an n -qubit state, one can write

$$|\psi\rangle = |0\rangle|\psi_0\rangle + |1\rangle|\psi_1\rangle.$$

Measuring the first qubit gives the corresponding classical outcome and leaves the remaining system in the appropriate conditional state.

14.2 Adding a Fresh Qubit

Given an n -qubit state $|\psi\rangle$, we can initialize a fresh qubit as $|0\rangle$, generating

$$|\psi\rangle \otimes |0\rangle,$$

a state on $n + 1$ qubits.

15 Joining and Separating Systems

More generally, we can join quantum systems:

$$|\psi\rangle \otimes |\phi\rangle.$$

We can also separate unentangled systems. These are not additional axioms; they can be derived from the axioms.

15.1 Why the Description Gives the Correct Statistics

Suppose

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Then

$$|\psi\rangle \otimes |\phi\rangle = \alpha|0\rangle \otimes |\phi\rangle + \beta|1\rangle \otimes |\phi\rangle.$$

If we measure the first qubit and obtain 1, the remaining state is proportional to $|\phi\rangle$, as expected for an unentangled system.

16 Important Math Review: Probability Theory

Let $X \geq 0$ be a random variable.

16.1 Markov's Inequality

For all $a > 0$,

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Also,

$$\mathbb{E}[X] = \sum_x x \Pr(X = x)$$

for a discrete random variable.

16.2 Chebyshev's Inequality

Let X be a random variable with finite variance

$$\text{Var}(X) = \sigma^2$$

and mean

$$\mathbb{E}[X] = \mu.$$

For $k > 0$,

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

This follows by applying Markov's inequality to

$$Y = (X - \mu)^2$$

with threshold $a = k^2\sigma^2$.

16.3 Chernoff Bounds

For a random variable X and suitable $t > 0$,

$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq e^{-ta} \mathbb{E}[e^{tX}],$$

by Markov's inequality.

If

$$X = X_1 + \cdots + X_n$$

for independent random variables, then

$$\mathbb{E}[e^{tX}] = \mathbb{E}\left[e^{t\sum_i X_i}\right] = \prod_i \mathbb{E}[e^{tX_i}].$$

16.4 Bernoulli Case

Let $X_i \in \{0, 1\}$ with

$$\Pr(X_i = 1) = p_i, \quad \Pr(X_i = 0) = 1 - p_i.$$

Then

$$\mathbb{E}[e^{tX_i}] = (1 - p_i) + p_i e^t = 1 + p_i(e^t - 1).$$

Using

$$1 + x \leq e^x,$$

we obtain

$$\mathbb{E}[e^{tX_i}] \leq \exp(p_i(e^t - 1)).$$

Therefore,

$$\mathbb{E}[e^{tX}] \leq \exp\left((e^t - 1) \sum_i p_i\right).$$

Since

$$\mu = \mathbb{E}[X] = \sum_i p_i,$$

we get

$$\mathbb{E}[e^{tX}] \leq e^{\mu(e^t - 1)}.$$

Consequently,

$$\Pr(X \geq a) \leq \inf_{t>0} \exp(-ta + \mu(e^t - 1)).$$

Taking

$$a = (1 + \delta)\mu$$

and optimizing gives

$$t = \ln(1 + \delta).$$

This yields the standard Chernoff upper-tail bound

$$\Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}}\right)^\mu.$$

A common simpler form is

$$\Pr(X \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2 + \delta}\right), \quad \delta > 0.$$

Similarly, for $0 < \delta < 1$,

$$\Pr(X \leq (1 - \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2}\right).$$

Exercise

Let

$$X = \sum_{i=1}^n X_i, \quad \mu = \mathbb{E}[X].$$

Derive the corresponding Chernoff bounds directly from Markov's inequality.